

HAT-P-32b

R-0266

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_250901 · created 2026-07-17T09:53:35+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460919.9198 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.053
Transit depth (Rp/Rs)^2	3.43 +/- 1.97 [%]
Orbital Inclination (inc)	89.55 +/- 1.73
Airmass coefficient 1 (a1)	0.82 +/- 0.14
Airmass coefficient 2 (a2)	0.28 +/- 1.6
Scatter in the residuals of the lightcurve fit is	18.76 %
Best Comparison Star	#2 - [590, 82]
Optimal Method	PSF photometry
Transit Duration (day)	0.0543 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.185 ± 0.053 vs 0.1489 (deviation 0.58σ)
Duration fit vs published	0.0543 d vs 0.1304 d (deviation 11.85σ)
Mid-time O-C	-2.6 min
Depth SNR	3.0
Residual scatter	18.76 %

Light curve

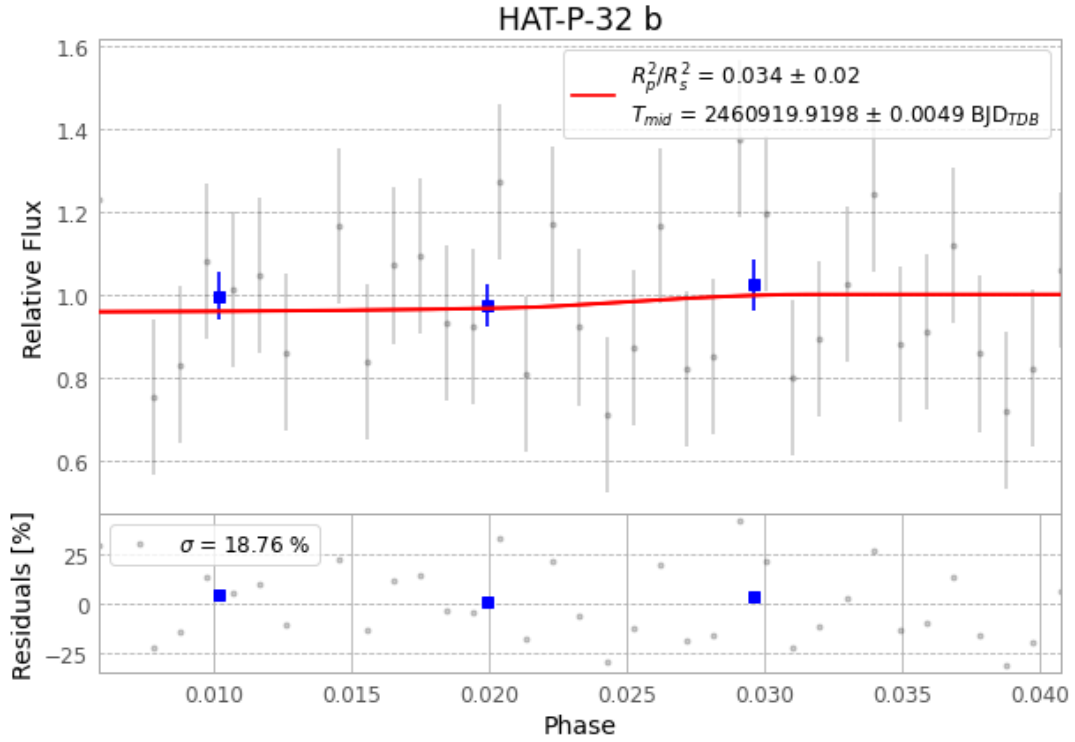


Figure 1 — FinalLightCurve_HAT-P-32 b_2025-09-01.png (SHA-256 3d767a4f4016ad18...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [63, 68], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7e661c48f2d232fb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 7dbc7ef

Data provenance

36 FITS frames (24 MB), HATP-32250901101814.FITS → HATP-32250901120618.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-250901.log (2ebb17866fdc...), FinalLightCurve_HAT-P-32 b_2025-09-01.png (3d767a4f4016...), FinalLightCurve_HAT-P-32 b_2025-09-01.csv (ba157ce54f85...)

Compute resources

Wall time 155.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 09:53Z.